Blueprint2CAD: Benchmarking Foundation Models on Engineering Drawing–to–CAD Render Correspondence

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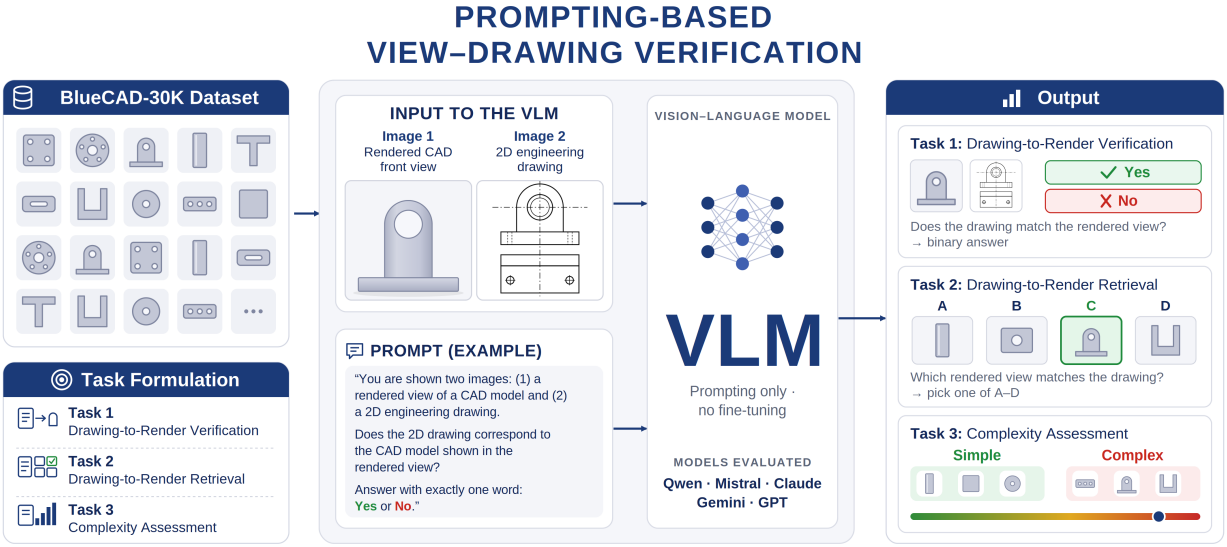


Figure 1. Overview of Blueprint2CAD. All tasks pair two 2D images, an engineering line drawing and a render of a part.

Abstract

Engineering drawings specify mechanical parts through orthographic projections. Recognizing which part a drawing depicts underlies part search, design reuse, and digitization, yet no public benchmark measures whether vision models can do it. We introduce ***Blueprint2CAD****, a benchmark for drawing-to-render correspondence: given an orthographic line drawing and a shaded render, usually from another viewpoint, decide whether they show the same part. Its data layer,* ***BlueCAD-30K****, applies an open drafting pipeline to 29,925 Zero-to-CAD models, adding view-aligned renders, clean and dimensioned drawings from six viewpoints, and surface-type metadata. Frozen encoders retrieve the matching part from a 10,000-model gallery with up to 65% Recall@1 when viewpoints are known but 43% when they are not, and the four encoders we test lie within six points of each other: viewpoint, not backbone, is the bottleneck. Six zero-shot vision-language models reach only 0.52–0.67 verification accuracy, with model-specific* Yes/No *biases that chain-of-thought shifts but does not remove; a quick LoRA fine-tuning on in-domain pairs improves verification accuracy by a few points but sharpens these biases and leaves the viewpoint failure intact.* [1]

1. Introduction

Engineering drawings are how mechanical parts are specified, archived, and exchanged. A drawing encodes shape implicitly, through orthographic projections and drafting conventions [22, 31], while the part itself exists as an exact CAD boundary representation (B-rep) [54, 67]. Recognizing which part a drawing depicts is the first step in part-library search, legacy-design reuse, drawing digitization, and model verification, and any automated pipeline

[1] Code and benchmark are available at: https://github.com/alexvalgui/blueprint2cad.

from drawings to solids depends on a visual front-end that does this reliably.

The problem is harder than it looks. Mechanical components are textureless, symmetric, and repetitive, and their identity rests on small features such as holes, pockets, fillets, and chamfers. A drawing presents these as sparse line art with hidden lines and annotations layered on top, whereas a render shows a shaded surface, usually from another viewpoint. Deciding whether the two show the same part therefore means holding one projection and relating it to another across a change of view, a correspondence that is natural to carry as visual structure and awkward to spell out in words. Existing CAD datasets [37, 39, 52, 82] support shape retrieval and generative modeling, and SPARE3D [25] poses spatial-reasoning questions over three-view drawings, but none pairs engineering drawings with shaded renders of the same part under controlled cameras, so the encoders behind drawing-to-CAD reconstruction [19, 81], drawing digitization [17, 58], and CAD inspection [1] have never been measured on orthographic line art.

We introduce **Blueprint2CAD**, a benchmark that makes drawing-to-render correspondence a first-class task. Its scope is deliberately 2D: every task relates a line drawing to an image of the part it describes, and nothing is claimed about recovering 3D geometry. The data layer, **BlueCAD-30K**, adds no new geometry either. It is the output of an open drafting pipeline applied to 29,925 models of the Zero-to-CAD collection [5]: shaded renders together with clean and dimensioned projections from six canonical viewpoints under a shared camera rig, so that view correspondence holds by construction, plus B-rep-derived surface-type metadata and model-level training, validation, and test splits. The pipeline runs on any STEP collection, which is how we expect the benchmark to grow.

We evaluate the two paradigms a practitioner would reach for first. Five frozen vision foundation models (DINOv3 [72], SigLIP2 [75], I-JEPA [4], RADIO [66], TIPSv2 [11]) serve as embeddings for search over a 10,000-model gallery: when the camera direction of each projection is known, the four strongest reach 59–65% Recall@1 within six points of each other, while a hand-crafted silhouette descriptor already reaches 49%; without that knowledge the best encoder falls to 43%, still about twice the silhouette's rate, two dimension lines per view cost at most seven points, and I-JEPA stays near chance under our protocol. Six vision-language models (Claude Sonnet 5 [2], Gemini 3.6 Flash [23], GPT-5.6 Luna [60], Mistral Small 3.2 24B [56], InternVL3.5 14B [80], Qwen3-VL 8B [8]) are queried zero-shot on binary verification and 4-way retrieval: none reaches 0.70 verification accuracy, even with chain-of-thought, each carries its own *Yes*/*No* prior that reasoning prompts redirect rather than remove, and correct matches become rare once the render is a top or bottom view. A tiny LoRA adapter trained on in-domain pairs lifts Qwen3-VL 8B by a few points without touching either failure mode. Read together, the two studies place drawing-to-render correspondence among the problems where an intermediate representation in language does not stand in for one that keeps the geometry: supplying the view relation directly, as the oracle alignment does for frozen encoders, is worth more than any change of backbone, while asking a Vision Language Model (VLM) to narrate that relation leaves its viewpoint failure where it was. Our contributions are:

- **Blueprint2CAD**, a 2D benchmark for matching engineering drawings to part renders, covering gallery-scale retrieval, two direct VLM tasks, and an unguided complexity probe.
- An open drafting pipeline for any STEP collection, and **BlueCAD-30K**, the line art, shaded images, metadata, and model-level splits it adds to 29,925 Zero-to-CAD models.
- A **diagnostic study** of the five encoders against a non-learned silhouette baseline, under view alignment, multi-view fusion, and annotation clutter, and a failure-mode analysis of the six VLMs and one fine-tuned variant that isolates model-specific response biases and tests whether reasoning in language recovers the cross-view correspondence the task requires.

2. Related Work

Reasoning with visual intermediates. A model can externalize an intermediate as pixels or tokens [9, 29, 49, 73], keep it latent in the encoder's embedding space [47, 86, 94], or imagine the unseen view with a world model or spatial map [78, 93], because textual chain-of-thought does not improve and can degrade spatial judgments [34, 92], binary verification follows a model-specific *Yes* prior [12], and VLMs default to the camera's frame under viewpoint change [55, 97, 98]. None tests whether an intermediate carries an orthographic drawing to a render of the same part; Blueprint2CAD measures that, and its view labels make these methods testable on it.

Engineering drawings Drawings have long been treated as a document-analysis problem: vectorization of scans, symbol and dimension parsing [17, 58], and recently drawing-centric VQA benchmarks for multimodal models [41, 59], with reconstruction from orthographic views [28, 30, 63, 81] as the downstream goal. SPARE3D [25] comes closest, asking view-consistency and camera-pose questions over three-view line drawings, but both sides of each question are line art. Missing is a measurement of the step in between: whether a general-purpose

encoder or VLM can read a drawing well enough to recognize the part it depicts.

2D–3D CAD retrieval and foundation models. Image-to-CAD retrieval learns joint embeddings of images and 3D geometry, from RGB-D scans [7] to monocular RGB [21, 24, 42, 43, 53], and sketch-based retrieval addresses the line-drawing-to-shape gap on SHREC [14, 16, 45, 46, 64]; both target household objects with dedicated 3D encoders. Frozen encoders such as CLIP [65], DINO [61, 72], SAM [38], and RADIO [66] are distilled into 3D encoders by ULIP [90, 91], OpenShape [50], and Uni3D [99], but their natural-image pretraining is not tuned to untextured, symmetric mechanical parts; CADCrafter [13] conditions on depth and normals instead, and a growing line fine-tunes multimodal Large Language Models (LLMs) to emit parametric CAD directly [15, 68, 79, 85, 87, 95, 96]. We instead measure purely 2D encoders with a line drawing as the query and a render of an engineering part as the target.

VLMs and native CAD learning. VLM work on CAD covers generation from visual input [20, 33, 40], drawing VQA where correspondence is a small subtask [41, 59], and embedding-space sketch matching [26]; none poses identity verification between an engineering drawing and a shaded render as a primary task on a public dataset. Separately, generative models emit sketch-and-extrude sequences [84, 88, 89], B-rep encoders classify and retrieve parts [32, 44], and reverse engineering recovers models from point clouds [35, 51, 76], sketches [48, 70], or drawings [28, 63, 81], supported by datasets with design histories [36, 69, 82]. We use B-rep machinery only to describe and sample our data; B-rep-native encoders are left to future work.

3. Background

CAD systems [67, 74] describe a solid as a B-rep: faces, edges, and vertices carried by parametric surfaces and curves [10, 54], with freeform geometry expressed as NURBS [62] and built as sequences of operations over constrained 2D sketches [71]. The CAD model is the primary definition from which engineering drawings are derived. A drawing follows Monge's descriptive geometry [57]: a 3D object is represented exactly by its orthographic projections onto perpendicular planes, stored digitally as vector primitives (segments, arcs, circles, splines) interleaved with dimensions, text, and hatching in formats such as DXF [6] or SVG [83]. In practice drawings range from bare orthographic outlines to section and auxiliary views, GD&T, surface-finish callouts, and title blocks [22, 31]; this work addresses the orthographic core shared by all of them (Sec. 7). Because the 3D geometry is encoded implicitly across complementary views, reconstructing a model from its drawing has been an open problem since the earliest work on orthographic reconstruction [30].

4. BlueCAD-30K

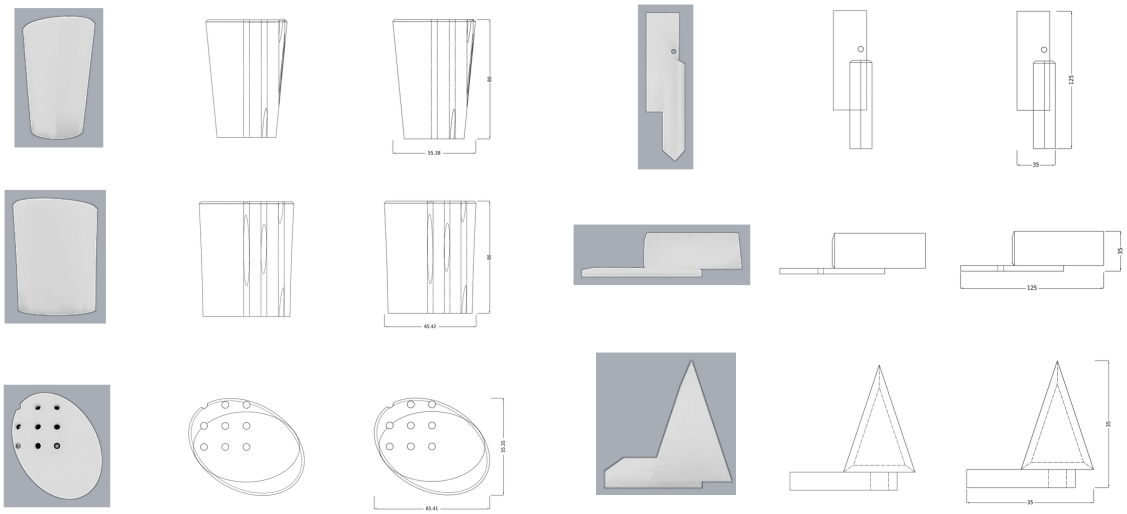
4.1. Dataset

Origin. BlueCAD-30K is built on Zero-to-CAD [5], a 2026 collection of STEP models generated from procedural CAD scripts. We add no geometry: the dataset is the drawings, renders, and metadata our pipeline produces for a filtered subset, with source models referenced by identifier. We chose Zero-to-CAD over scraped repositories [37, 39, 82] because paired drawing generation needs clean, watertight, single-solid geometry; the pipeline runs on any STEP collection and can be re-instantiated on real part libraries (Sec. 7).

Filtering. A B-rep parser removes models with a low geometry score or curved-surface ratio, degenerate aspect ratios, non-watertight geometry, or fewer than $k{=}5$ faces, reducing the 1M pool to roughly 300K. A random 10% sample gives 30,237 models; 29,925 pass every stage, and 312 (1.0%) fail in drafting or rendering because of invalid topology the parser misses. In the sample, 65% of faces are planar and 22% cylindrical, 24% of models contain a B-spline face, and 0.02% are exact topological duplicates; geometric diversity is bounded by the single generator (Sec. 7).

Drafting pipeline. Each model is projected in native STEP coordinates from six canonical orthographic viewpoints shared by drawing and render, so view correspondence holds by construction (Fig. 3; axes and edge treatment in the supplementary). Drawings use OpenCascade hidden-line removal, are stored as layered DXF, and are rasterized to 1024×1024 px on white; renders have the same resolution and a transparent background. A dimensioned variant adds width and height dimensions to each view. Data (CC BY 4.0) and code (Apache 2.0) are released: per model, six renders, six clean and six dimensioned drawings, and B-rep metadata, plus the VLM pairs, prompts, checkpoints, and raw responses.

Splits. Splits are made by model identifier, so no model appears in two of them: 27,479 models for training, 1,446 for validation (5% of the non-test models, fixed seed), and 1,000 for testing, which form the VLM evaluation set (Sec. 4.2). Training pairs draw negatives only from their own split (supplementary Sec. D). The frozen-encoder study uses a 10,000-model gallery sampled from the dataset. All splits, including the gallery, are released.



(a) Render (b) Drawing (c) Drawing with dimension (a) Render (b) Drawing (c) Drawing with dimension

Freeform subset (*complex*) Prismatic subset (*simple*)

Figure 2. **Examples of CAD models** from the two VLM evaluation subsets: shaded render, clean 2D drawing, and dimensioned 2D drawing.

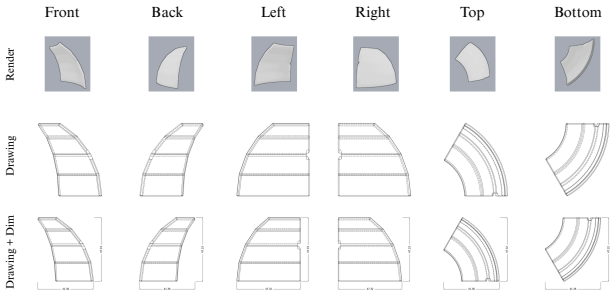


Figure 3. **A CAD example** with its six camera directions and paired render, drawing, and dimensioned drawing views.

4.2. VLM Evaluation Set

The VLM evaluation set holds the 500 lowest- and 500 highest-scoring models under a selection score $s \in [0, 100]$, a weighted sum of seven normalized B-rep signals (freeform-face ratio, face count, non-planar ratio, surface-type entropy, inner loops, curved-edge ratio, and sphericity; weights in supplementary Table 8, following [3, 32, 44, 77]). The low end consists of *prismatic* parts with planar and cylindrical faces, the high end of *freeform* parts dominated by B-spline surfaces. The score only selects the two sets, which do not overlap in score (12.2–18.2 vs. 59.1–87.3): 99% of prismatic models have no freeform face and 95% of freeform models have a non-planar ratio above 0.6, while face counts overlap (interquartile ranges 19–32 and 26–63), so the classes differ in surface type, not size (supplementary Sec. A). Tables label them *simple* and *complex* to match the Task 3 prompt.

5. Blueprint2CAD Benchmark

We examine drawing-to-render correspondence from two angles: frozen Vision Foundation Model (VFM)s as cross-modal embeddings for retrieval (Sec. 5.1), and VLMs queried directly on matching tasks (Sec. 5.2). Retrieval runs over a 10,000-model sample of the full dataset, and the VLM tasks over the 1,000-model evaluation set. Every input is a pair of 2D images, a line drawing on one side and a shaded render on the other. Models run frozen or zero-shot, with one exception, a LoRA-adapted Qwen3-VL 8B (Sec. 5.2.1), a small-scale reference that exercises the training split (Sec. 7).

5.1. Vision Foundation Models

We ask whether frozen 2D VFMs can link a drawing to the render of the same part, probing four backbones with distinct pretraining objectives: DINOv3 [72], SigLIP2 [75], RADIO [66], and TIPSv2 [11]. Freezing is deliberate: digitization and reconstruction pipelines use these encoders off the shelf, so we measure what pretraining already captures. Since nothing is learned, no training split is needed, and the gallery is a 10,000-model sample of the dataset. Each backbone yields one global descriptor per drawing and render view of every model, twelve in total (Sec. 4.1), mean-centered per modality over the gallery and ℓ_2-normalized; checkpoints, resolutions, and descriptor choices are in the

supplementary.

Cross-modal retrieval protocol. Each model thus has $V{=}6$ unit-normalized drawing descriptors $\hat{d}_i$ and render descriptors $\hat{r}_i$, indexed by viewpoint. A query in one modality must retrieve its counterpart among all 10,000 gallery models; we report bidirectional Recall@k (R@k), mean reciprocal rank (MRR), and median rank (chance: R@1 $=$ 0.01%, median rank $\approx$ 5,000). With unknown viewpoints (*view-agnostic*), *Max* takes the highest similarity over all $V{\times}V$ pairs and *Mean* compares the renormalized per-model averages. With known viewpoints (*view-aligned*), only matching projections are compared, $s_{\text{aligned}} = \text{agg}_i\ \hat{d}_i^{\top}\hat{r}_i$, with agg the mean (*Al-mean*) or the max (*Al-max*). Since alignment uses oracle labels, it is a diagnostic of what a representation can do once viewpoint is solved; the view-agnostic setting is the deployable one.

Experiment 1: Frozen Backbones. Tab. 1 separates two factors. The backbone matters little: DINOv3, TIPSv2, SigLIP2, and RADIO, pretrained by self-distillation, image–text contrast, and multi-teacher distillation, finish within six R@1 points of each other under Al-mean. Viewpoint correspondence matters far more, adding 22–30 points over each encoder's best view-agnostic score, so the deployable setting is limited by viewpoint rather than by representation quality. Outline alone explains much of this: the silhouette baseline reaches three quarters of the encoders' aligned R@1 and gains as much from alignment. The encoders lead it by 11–17 points when viewpoints are known but retrieve 1.7–2.1 times as often when they are not, so pretraining mainly adds a view-consistent description of the part rather than a reading of its interior. Averaging over views beats the best single view in both settings, since several weak projections outweigh one that different parts may share. All four encoders retrieve better from drawings to renders than the reverse while the silhouette baseline is symmetric, placing the asymmetry in how the encoders embed line art and shaded renders. DINOv3 has the best aligned R@1 and MRR and is the default below.

Experiment 2: Per-View Analysis. Tab. 2 adds views to DINOv3 one at a time. Gains come from new viewing axes, not from more pictures: *back* adds under a point to *front*, and for all four encoders an opposing pair improves on its stronger member by at most 2.5 points (supplementary), since the two projections mirror each other. Plan views carry the most information: *top* and *bottom* are the strongest single views for every encoder, roughly twice any side view, and top + left is every encoder's best pair, plausibly because extruded mechanical parts expose holes and pockets in plan that side views reduce to lines. The late jump at *top* follows from this ranking, as it outweighs the weaker *bottom* view it mirrors; the sixth view then changes R@1 by under half a point for every encoder.

Table 1. **Experiment 1: frozen-backbone cross-modal retrieval.** R@1 (%, D→R) for the four scoring strategies defined in the protocol: *Max*/*Mean* are view-agnostic, *Al-max*/*Al-mean* assume known viewpoints. MRR, median rank, and reverse direction for Al-mean. *Silhouette*: non-learned outline baseline, six filled 16×16 view masks plus aspect ratio (the hard-negative descriptor of Sec. D). Bold: best per column. Full metrics in the supplementary.

	R@1, view-agnostic ↑		**R@1**, view-aligned ↑		**Al-mean**		
Backbone	**Max**	**Mean**	**Al-max**	**Al-mean**	**R@1 (R→D)** ↑	**MRR** ↑	**MedR** ↓
Silhouette	7.85	20.16	10.96	48.54	48.08	0.555	2
DINOv3	29.10	**43.11**	34.67	**65.14**	57.57	**0.728**	**1**
TIPSv2	**34.39**	34.65	**38.58**	64.65	**58.67**	0.724	**1**
SigLIP2	29.52	38.07	33.54	64.15	57.15	0.714	**1**
RADIO	27.46	36.54	30.98	59.51	49.58	0.675	**1**

Table 2. **Experiment 2: cumulative view fusion** (frozen DINOv3, Al-mean, D→R). Gains concentrate where a new viewing axis is introduced and on the top view; the opposing back and right views add almost nothing. Single- and two-view results for all encoders in the supplementary.

Views	**R@1**↑	**R@5**↑	**R@10**↑	**MRR**↑	**MedR**↓
front	14.88	25.98	31.38	0.206	62
+ back	15.76	27.08	33.11	0.217	48
+ bottom	42.19	61.12	68.20	0.512	2
+ left	52.15	70.74	76.93	0.607	**1**
+ top	**65.59**	**81.90**	**87.15**	**0.731**	**1**
+ right (all 6)	65.14	81.79	86.87	0.728	**1**

Table 3. **Experiment 3: annotation ablation** (normal → dim, Al-mean, D→R). Worse / Better: share of queries whose rank degrades / improves with dimension annotations; the rest keep their rank. Bold: best value before and after annotation. Full metrics in the supplementary.

Backbone	**R@1** (%) ↑	**MRR** ↑	**MedR**	**Worse** (%)	**Better** (%)
DINOv3	**65.14** → 58.10	**0.728** → 0.663	1 → 1	29.1	13.4
TIPSv2	64.65 → **61.61**	0.724 → **0.697**	1 → 1	23.2	16.0
SigLIP2	64.15 → 61.51	0.714 → 0.692	1 → 1	23.7	20.3
RADIO	59.51 → 58.84	0.675 → 0.673	1 → 1	22.5	22.9

Experiment 3: Drawing Annotations. Real engineering drawings carry dimension annotations (extension lines, arrows, measurement text) with no counterpart in a render. Our dimensioned variant (Tab. 3) adds two callouts per view, far fewer than a production drawing carries, and its cost is small next to the viewpoint problem: at most 7 R@1 points against 22–30 for unknown viewpoints. Annotations reorder close candidates rather than lose the match: more than half of the queries keep their exact rank, the rest move by a median of 4–5 positions, and every gap narrows from R@1 to R@10 (supplementary). Sensitivity differs by backbone. DINOv3, the only one of the four trained with-

out language supervision, loses most and drops from first to last, while RADIO's change is not significant (sign test $|z| = 0.5$). With four encoders this link to language supervision is a correlation, and whether the measurement text or the extension lines drive it remains untested.

5.2. Vision-Language Models Benchmark

We evaluate six VLMs, three closed-source (Gemini 3.6 Flash [23], GPT-5.6 Luna [60], Claude Sonnet 5 [2]) and three open-source (Mistral Small 3.2 24B [56], Qwen3-VL 8B [8], InternVL3.5 14B [80]), zero-shot on the 1,000-model evaluation set of Sec. 4.2 with fixed, untuned prompts and exact string matching; model identifiers and decoding and reasoning settings are listed in supplementary Tab. 13, and pairs, views, prompts, and raw responses are released. A seventh entry, Qwen3-VL 8B LoRA-tuned on Task 1 training pairs (supplementary Sec. D), uses the same harness and is listed as *Qwen3-VL 8B (FT)*; bold marks compare zero-shot models only.

Task 1: Drawing–Render Verification. Models judge whether a rendered view and a 2D drawing depict the same part:

VLM Prompt

You are shown a rendered view of a CAD model and a 2D engineering drawing. The two images may show the part from different viewpoints. Do they depict the same part? Answer with exactly one word: Yes or No.

The drawing is always the front view; the render is a uniformly sampled canonical view of the same part (50%) or of a different one (50%), the latter taken from the whole evaluation set (25%) or from the same surface-type class (25%). In five of six positive cases the render viewpoint differs from the drawing, so silhouette matching fails even on true positives. Dimensioned and undimensioned drawings are evaluated separately. A control tests whether the forced-choice format suppresses reasoning and, more to the point, whether making the view change explicit in language helps: every model is also run on the undimensioned pairs with a chain-of-thought (CoT) variant that asks it to first think step by step about silhouette, proportions, and distinctive details (holes, slots, steps), and about how the drawing would look from the render's viewpoint, then answer Yes or No on a new final line; only that line is scored, and unparseable responses are excluded and counted.

Task 2: Drawing-to-Render Retrieval. Given the right-view drawing of a part, models pick the matching part among four randomly ordered right-view renders, one correct and three distractors sampled uniformly from the evaluation set (chance = 0.25). Distractor difficulty is not controlled, so some items are easy; the breakdown by surface-type class is in the supplementary. This is a small-gallery analogue of Sec. 5.1: can a VLM do explicitly what frozen features do through embedding similarity?

VLM Prompt

The first image is a 2D engineering drawing of a CAD model. The following 4 images are rendered CAD models, labeled A, B, C, D. Which rendered model does the drawing represent? Answer with exactly one letter.

Task 3: Unguided Complexity Probe. Given all six orthographic views, models label the part *Simple* or *Complex*. The prompt deliberately gives no criterion: the task probes whether a VLM's native notion of visual complexity tracks the prismatic/freeform distinction of Sec. 4.2, whose defining signal (surface type) is only indirectly visible in projection. It is diagnostic rather than a capability measure.

VLM Prompt

You are shown six views (front, back, left, right, top, bottom) of a CAD model. Classify the model as Simple or Complex based on its geometry. Answer with exactly one word: Simple or Complex.

5.2.1. Empirical Findings.

Verification. Table 4 summarizes all tasks. On Task 1 the six zero-shot models score 0.52–0.67 with or without dimensions, Per-model 95% intervals are about ± 0.03 at n=1000, but since every model answers the same items we compare models with paired exact McNemar tests on the items where their answers differ, Holm-corrected within each family of comparisons (supplementary Tab. 14); no pair among the four leading models differs in either condition ($p_{\text{Holm}} \geq 0.28$). Table 6 shows why: every model discriminates above chance, but a model-specific decision bias dominates its errors. Claude and GPT-5.6 lean toward `Yes`; Gemini, Mistral-S, and InternVL3.5 toward `No`; GPT-5.6 Luna is the most balanced ($|\Delta| = 0.066$). Dimensions push five of the six further toward `Yes` without a gain in accuracy; the only significant change is a loss, Claude's 4.5 points ($p_{\text{Holm}} = 0.020$; supplementary Tab. 14).

Errors also depend on viewpoint: positive accuracy peaks on side renders, which rotate about the vertical axis, and except for Gemini is lowest on top and bottom; Claude falls from 0.98 (right) to 0.46 (top) without dimensions (supplementary). Chain-of-thought does not help (Table 7): under a paired McNemar test of each model's CoT and forced-choice answers on the same items, no model gains significantly (largest gain Claude, +3.8 points, $p_{\text{Holm}} = 0.17$), the best CoT result (0.681, GPT-5.6 Luna) is +0.7

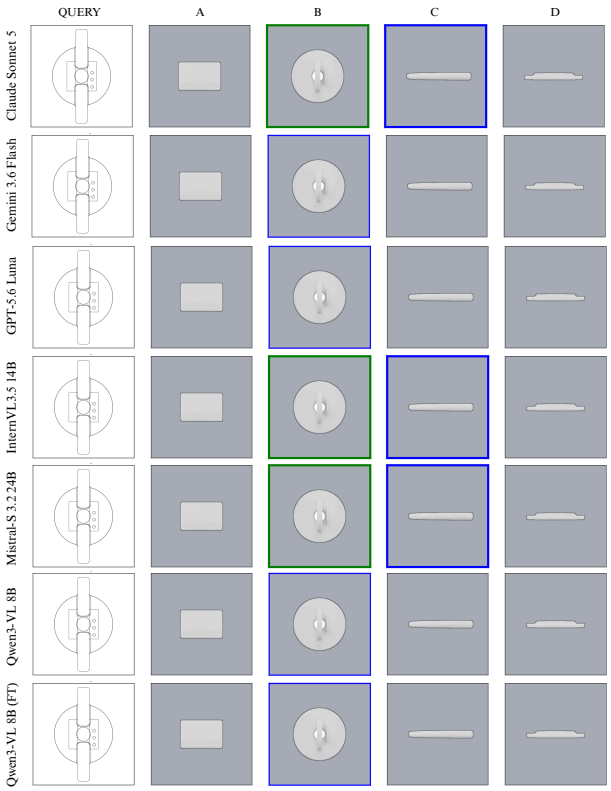


Figure 4. **Examples of 2D drawing retrieval tasks with four candidate CAD renders.** Green border: ground-truth render; blue border: model selection. A correct selection makes the two borders coincide.

points over its forced-choice answers ($p_{\text{Holm}} = 1.0$), and Qwen3-VL loses 4.9 points ($p_{\text{Holm}} = 0.007$; supplementary Tab. 14). CoT instead shifts each bias (Claude and GPT-5.6 flip toward rejecting, Mistral-S toward accepting), so the textual intermediate relocates the prior rather than supplying the view relation the task needs. The fine-tuned Qwen3-VL ignores the CoT instruction and reproduces its forced-choice predictions.

Retrieval. Gemini leads Task 2 by a wide margin (6.3 points over GPT-5.6 Luna on the items both answered, paired McNemar $p < 0.001$), GPT-5.6 Luna is the only other model far above chance, and Claude and Qwen3-VL are only modestly above it (supplementary), yet Gemini is mid-pack on Task 1: relative comparison appears easier than an absolute judgment. Open-source models stay near chance here (InternVL3.5 and Mistral-S are indistinguishable from it, Qwen3-VL is 7 points above) while equalling the closed-source leaders on Task 1, so their gap is specific to choosing among candidates.

Table 4. **Main VLM results (accuracy, macro-F1)**. Task 1: verification without and with dimensions (DIM), n=1000 for zero-shot models (one item lacks its render; supplementary Sec. C.1) and 1,000 for FT; per-model 95% Wilson intervals are at most $\pm$0.031, and paired model comparisons are in supplementary Tab. 14. Task 2: four-way retrieval (chance 0.25). Task 3: unguided complexity probe scored against surface-type class (chance 0.50). Bold: best model per column, not including the fine-tuned (FT) row.

	Task 1		Task 1 + DIM		Task 2		Task 3	
Model	Acc. ↑	F1 ↑	Acc. ↑	F1 ↑	Acc. ↑	F1 ↑	Acc. ↑	F1 ↑
Closed-source models								
Claude Sonnet 5	0.635	0.631	0.590	0.560	0.374	0.372	0.520	0.409
Gemini 3.6 Flash	0.654	0.620	**0.670**	0.646	**0.690**	**0.689**	0.562	0.491
GPT-5.6 Luna	**0.670**	**0.670**	0.654	0.649	0.637	0.636	**0.676**	**0.676**
Open-source models								
Mistral-S 3.2 24B	0.534	0.417	0.524	0.450	0.274	0.235	0.521	0.409
InternVL3.5 14B	0.624	0.604	0.637	0.618	0.266	0.274	0.506	0.399
Qwen3-VL 8B	**0.670**	0.669	0.660	**0.655**	0.317	0.305	0.512	0.363
Qwen3-VL 8B (FT)	0.685	0.665	0.699	0.688	0.241	0.246	0.553	0.491

Complexity probe. On Task 3 (Table 5), every model except GPT-5.6 Luna answers *Simple* on 88–98% of a balanced set, so near-chance accuracy comes almost entirely from that one label. Asked without a criterion, these models' notion of complexity does not follow surface type: a part dominated by B-spline faces can present as a plain silhouette in projection. GPT-5.6 Luna balances the two per-class recalls, so the signal is recoverable from the same input. In addition, the small Qwen3-VL 8B finetune shows interesting results on this task even without being trained at all on it.

Fine-tuning. Rank-16 LoRA adapters on the language model and vision encoder (52.5M parameters, 0.60%; Table 21), trained on 20,000 Task 1 pairs, raise Qwen3-VL 8B from 0.670 to 0.685 without dimensions, which is not significant (paired McNemar $p_{\text{Holm}} = 0.39$), and from 0.660 to 0.699 with them ($p_{\text{Holm}} = 0.031$). The gain comes from negatives: the adapted model rejects 0.930 of negative pairs (zero-shot 0.729) but accepts only 0.440 of positives (0.610), so its lean toward `No` grows from -0.119 to -0.490 (Table 6), while top and bottom renders stay at 0.14 and 0.19 on positives. On Task 2 it drops to chance (0.241, 7.6 points below zero-shot, $p_{\text{Holm}} < 0.001$); on Task 3 its *Simple* rate falls from 98.4% to 84.9% (accuracy 0.512 to 0.553, $p_{\text{Holm}} < 0.001$). At this scale, in-domain data buys a few points by shifting the threshold toward `No`, as its silhouette-mined negatives (70% of pairs, Sec. D) predict; whether more data or test-matched negatives change this is open.

6. Discussion

The two paradigms fail in different ways. Retrieval is a relative task, and frozen encoders handle it once view

Table 5. **Complexity probe (Task 3)**: per-class recall, overall accuracy and macro-F1. Labels are B-rep-derived (prismatic = Simple, freeform = Complex); the task probes alignment between an unguided notion of complexity and surface type.

Model	Acc. ↑	Macro F1 ↑	`Complex` Rec. ↑	`Simple` Rec. ↑
Closed-source models				
Claude Sonnet 5	0.520	0.409	0.086	0.954
Gemini 3.6 Flash	0.562	0.491	0.188	0.940
GPT-5.6 Luna	**0.676**	**0.676**	0.712	0.640
Open-source models				
Mistral-S 3.2 24B	0.521	0.409	0.087	0.954
InternVL3.5 14B	0.506	0.399	0.084	0.928
Qwen3-VL 8B	0.512	0.363	0.028	0.996
Qwen3-VL 8B (FT)	0.553	0.491	0.204	0.902

Table 6. **Task 1 response bias:** accuracy on positive (n=500) and negative (n=499; 500 for FT) pairs, without and with dimensions. Δ = Pos. − Neg.; Δ_{DIM} = change in Pos. from adding dimensions. Bold: best zero-shot model per column and smallest $|\Delta|$.

	Undimensioned			Dimensioned			
Model	Pos. ↑	Neg. ↑	Δ	Pos. ↑	Neg. ↑	Δ	Δ_{DIM}
Closed-source models							
Claude Sonnet 5	**0.728**	0.541	+0.187	**0.848**	0.331	+0.517	+0.120
Gemini 3.6 Flash	0.356	0.952	−0.596	0.412	**0.928**	−0.516	+0.056
GPT-5.6 Luna	0.703	0.637	**+0.066**	0.772	0.535	**+0.237**	+0.069
Open-source models							
Mistral-S 3.2 24B	0.086	**0.982**	−0.896	0.158	0.890	−0.732	+0.072
InternVL3.5 14B	0.400	0.848	−0.448	0.416	0.858	−0.442	+0.016
Qwen3-VL 8B	0.610	0.729	−0.119	0.540	0.780	−0.240	−0.070
Qwen3-VL 8B (FT)	0.440	0.930	−0.490	0.510	0.888	−0.378	+0.070

Table 7. **Task 1 under chain-of-thought prompting (undimensioned).** Accuracy, macro-F1, positive- and negative-pair accuracy, and Δ = Pos. − Neg., with the forced-choice Δ from Table 6 in parentheses. n: items with a parseable final `Yes`/`No`. Bold: best zero-shot model per column and smallest $|\Delta|$.

Model	Acc. ↑	F1 ↑	Pos. ↑	Neg. ↑	Δ (forced-choice)	n
Closed-source models						
Claude Sonnet 5	0.673	0.664	0.515	0.830	−0.315 (+0.187)	996
Gemini 3.6 Flash	0.667	0.639	0.386	**0.949**	−0.563 (−0.596)	989
GPT-5.6 Luna	**0.681**	**0.679**	0.604	0.759	**−0.155** (+0.066)	959
Open-source models						
Mistral-S 3.2 24B	0.545	0.523	**0.758**	0.331	+0.427 (−0.896)	1000
InternVL3.5 14B	0.617	0.577	0.312	0.922	−0.610 (−0.448)	1000
Qwen3-VL 8B	0.621	0.610	0.452	0.790	−0.338 (−0.119)	1000
Qwen3-VL 8B (FT)	0.685	0.665	0.440	0.930	−0.490 (−0.490)	1000

correspondence is given: objectives as different as self-distillation, image–text contrast, and multi-teacher distillation converge, and matching six outlines already comes within 11–17 R@1 points of them. Their advantage grows without viewpoints, where they retrieve about twice as often, so what pretraining contributes is a description of the part that stays consistent across views. VLMs face the harder absolute task of verifying a single pair, and their errors are systematic: each applies its own *Yes*/*No* prior, reasoning prompts move that prior without correcting it, a basic fine-tune sharpens it, and top and bottom renders defeat every model, so the bottleneck is the relation between views rather than the detail within them. Handing an encoder the view correspondence is worth more than any change of backbone, while asking a VLM to spell it out in words only moves its decision threshold; the task appears to need an intermediate state that keeps the part's geometry and can be re-projected, which neither paradigm carries and the released view labels make testable.

7. Scope and Limitations

Blueprint2CAD compares line drawings with 2D renders and makes no claim about 3D CAD understanding. Its drawings are clean orthographic projections from a single procedural generator, without section views, GD&T, title blocks, or scan artifacts; the released pipeline rebuilds the benchmark from other STEP collections, and real documents are the next test. View-aligned scoring uses oracle viewpoint labels and is a diagnostic beside the deployable view-agnostic setting. Evaluation uses frozen weights or zero-shot prompting plus one fine-tuned variant, and the prismatic/freeform classes follow B-rep surface type rather than expert judgments of complexity. Section views, complex drawing collections, and larger-scale fine-tuning are the natural next steps for this benchmark. We based this analysis completely on [5] dataset, future work could focus on extending this paper to other 3D CAD datasets such as [39, 41, 82] that could have different characteristics.

8. Conclusion

Blueprint2CAD makes drawing-to-render correspondence measurable, and BlueCAD-30K supplies the aligned data it needs. Frozen encoders retrieve parts at gallery scale once view correspondence is given, largely independent of pretraining objective, and unknown viewpoints cost far more than dimension lines. VLMs verify pairs through a label prior that chain-of-thought shifts but does not remove, and fail on top and bottom renders even when they discriminate well at a fixed view; LoRA fine-tuning leaves both in place. The bottleneck is neither resolution nor backbone but the relation between views, and whether a geometry-preserving textual intermediate closes it remains open. Section views, larger drawing collections, and larger-scale fine-tuning are natural next steps, and the released pairs and view labels make each testable.

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Blueprint2CAD: Benchmarking Foundation Models on Engineering Drawing–to–CAD Render Correspondence

Supplementary Material

A. Dataset Statistics

Geometric statistics of the 1,000-model VLM evaluation set and of the full BlueCAD-30K sample (30,237 models, including the 312 that failed rendering), computed from each STEP B-rep with the OpenCascade extractor behind the selection score (Table 8). *Simple* and *complex* denote the prismatic and freeform subsets, as in the main text.

Face-count threshold. The geometric filter removes models with fewer than k=5 faces: below that the unfiltered pool is almost entirely single extrusions of convex polygons, whereas 5–15 faces already covers stepped blocks, flanged plates, and parts with through holes. Over the full sample the smallest model has 7 faces and 4.1% have fewer than 15; in the VLM set the share is 4.4% (0.8% of the simple subset, 8.0% of the complex one, mostly B-spline-dominated parts). Every model is a single solid.

Sample composition. Over the full dataset the score spans 11.9–88.6 (median 33.5, interquartile range 26.6–41.9), so most data sits between the two VLM classes. 76% of models contain no freeform face; 65% of all faces are planar and 22% cylindrical, while B-spline faces make up 4.9% of the sample but 51% of the complex subset.

Class separation. The score ranges 12.2–18.2 for the simple subset and 59.1–87.3 for the complex one (medians 16.8 and 68.5) with no overlap, and the classes are separated on every weighted signal, not only on their sum: 99% of simple models have no freeform face and all have a non-planar face ratio below 0.2, whereas 95% of complex models exceed 0.6 and 99.6% have a curved-edge ratio above 0.5. Face count does not separate them (interquartile ranges 19–32 and 26–63 overlap), nor do sphericity and aspect ratio, so the distinction is one of surface type and topology rather than of size or silhouette.

Near-duplicates and validity. Grouping models by an exact topological signature (face, edge, and inner-loop counts, the 11-bin surface-type histogram, and bounding-box proportions to four decimals) finds no shared signature in the VLM set and only three pairs (0.02%) in the full sample, so exact duplicates do not affect retrieval.

Edge treatment. Drawings use the `HLRBRep_HLRToShape` edge sets: sharp (`V/HCompound`) and silhouette (`OutLineV/HCompound`) edges are drawn solid when visible and dashed when hidden, while tangent, seam, and isoparametric edges are suppressed, matching the TechDraw default.

Table 8. Selection-score signals, weights (sum to 100), and normalization to $[0, 1]$. r: raw ratio for the named signal; H: Shannon entropy of the surface-type histogram, normalized by $\log(11)$; c: Wadell's sphericity, $c = \sqrt[3]{36\pi V^2}/A$, clipped to $[0, 1]$. Weights are fixed manually: the freeform-face ratio as the clearest primitive-vs-organic separator, surface-type entropy after the information-theoretic measures of SHREC'21 [3] with the per-face classification of [32, 44], and compactness as Wadell's sphericity [77] entered as $1 - c$.

Signal	Weight	Normalization
Freeform face ratio	25	$\min(1,\ 2r_{\text{freeform}})$
Face count (log)	15	$\min(1,\ \log_{10}(\text{faces})/3)$
Non-planar face ratio	15	$r_{\text{curved face}}$
Surface-type entropy	15	$\min(1,\ 1.5\,H)$
Inner boundary loops	12	$\min(1,\ \text{loops}/12)$
Curved edge ratio	10	$r_{\text{curved edge}}$
Shape compactness	8	$1 - c$

Table 9. Per-model medians for the VLM subsets (Simple = prismatic, Complex = freeform) and the full sample (n=30,237); interquartile ranges are visible in Fig. 5. Bottom block: fraction of all faces per surface class.

	Simple	Complex	Full
Faces	23	42	37
Edges	120	209	190
Inner loops	0	4	4
Freeform face ratio	0.00	0.50	0.00
Non-planar face ratio	0.07	0.81	0.30
Curved edge ratio	0.08	0.86	0.28
Surface-type entropy	0.11	0.43	0.28
Sphericity c	0.50	0.43	0.41
Bbox aspect	5.0	2.2	4.2
Selection score s	16.8	68.5	33.5
Plane faces	0.92	0.19	0.65
Cylinder faces	0.07	0.19	0.22
B-spline faces	0.00	0.51	0.05
Other faces	0.01	0.11	0.08

B. Frozen-Backbone Results

Checkpoints. DINOv3 ViT-H+/16, SigLIP2 So400m/14 (384 px), RADIO ViT-H/16, and TIPSv2 ViT-g/14 (448 px).

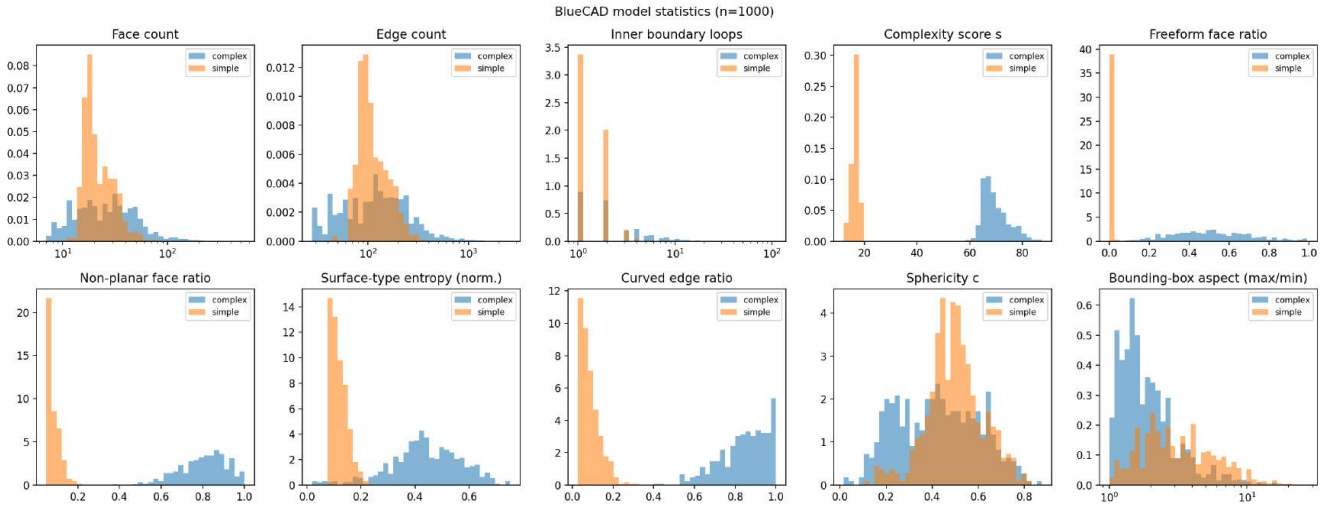


Figure 5. Distributions of B-rep signals for the VLM evaluation subsets (simple in orange, complex in blue) against the full sample (grey). Face count, edge count, and aspect ratio are on a log axis.

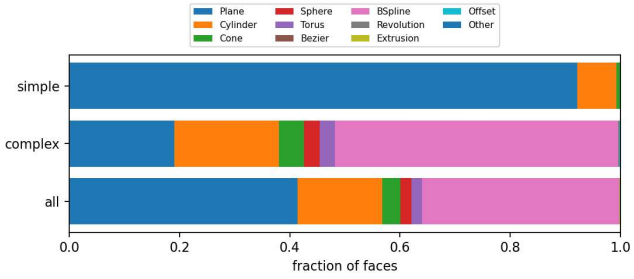


Figure 6. Fraction of faces per surface type for the simple subset, the complex subset, and the full sample. Simple parts are almost entirely planar and cylindrical; complex parts are dominated by B-spline faces.

Table 10. Canonical camera definitions. Projection directions point from the part toward the viewer (FreeCAD convention); up = direction $\times$ x-axis. All vectors are in native STEP coordinates, with no per-model reorientation.

View	Proj. direction	Image x-axis	Image up
Front	$(0,-1,0)$	$(1,0,0)$	$(0,0,1)$
Right	$(1,0,0)$	$(0,1,0)$	$(0,0,1)$
Back	$(0,1,0)$	$(-1,0,0)$	$(0,0,1)$
Left	$(-1,0,0)$	$(0,-1,0)$	$(0,0,1)$
Top	$(0,0,1)$	$(1,0,0)$	$(0,1,0)$
Bottom	$(0,0,-1)$	$(1,0,0)$	$(0,-1,0)$

HF models use each checkpoint's default image processor (resize and center-crop); TIPSv2 images are resized directly to square. Renders are stored as RGBA with a transparent background and converted to RGB before encoding, so every encoder sees them on a black background, whereas drawings are on white. Global descriptors are the CLS token (DINOv3, TIPSv2), the pooled output (SigLIP2), or the summary token (RADIO), computed in fp16 autocast, mean-centered per modality over the gallery, and ℓ_2-normalized. The *Silhouette* rows use the model-free descriptor of Sec. D: the foreground mask of each view (drawings: outline closed and filled, so hidden and interior lines are discarded), cropped to its bounding box, resampled to 16×16, concatenated with log aspect ratio and fill fraction, and ℓ_2-normalized.

B.1. Full-Gallery Retrieval

Tab. 11 extends Tab. 1 with all recall cutoffs, MRR, and median rank in both retrieval directions for the view-aligned (Al-mean) setting. All four encoders place the correct part first for most queries and within the top ten for 82–87% of them. Each retrieves better from drawings to renders than the reverse at every cutoff, by the widest margin for RADIO, whereas the outline baseline is symmetric.

B.2. Per-View Analysis

Tab. 12 gives single- and two-view results for all four encoders, and the pattern is the same for all of them. An opposing pair (front + back, left + right, top + bottom) improves on its stronger member by at most 2.5 R@1 points, whereas an orthogonal pair adds 11–22 points; *top* is the strongest single view and *bottom* the second, and top + left is the best pair. Information is thus spread across axes rather than across views, and the plan view carries the most of it.

B.3. Annotation Ablation

Every gap between clean and dimensioned drawings narrows from R@1 to R@10 (DINOv3 7.0 to 5.3 points,

Table 11. **Complete frozen-backbone retrieval results** (extends Tab. 1). View-aligned setting (Al-mean, viewpoints known), both directions, all metrics; *Silhouette* is the non-learned outline baseline. Gallery: N=10,000-model sample; chance R@1 = 0.01%, chance median rank ≈ 5,000.

Backbone	Direction	R@1 ↑	R@2 ↑	R@3 ↑	R@5 ↑	R@10 ↑	MRR ↑	MedR ↓
Silhouette	D→R	48.54	55.41	59.04	62.79	68.34	0.555	2
	R→D	48.08	55.40	58.75	63.14	68.98	0.554	2
DINOv3	D→R	65.14	73.56	77.67	81.79	86.87	0.728	1
	R→D	57.57	66.34	70.32	75.28	81.44	0.659	1
TIPSv2	D→R	64.65	73.34	77.08	81.63	86.42	0.724	1
	R→D	58.67	66.60	71.11	75.55	81.18	0.665	1
SigLIP2	D→R	64.15	71.99	75.50	79.71	84.50	0.714	1
	R→D	57.15	65.17	69.29	73.61	79.27	0.649	1
RADIO	D→R	59.51	67.86	72.37	77.01	82.36	0.675	1
	R→D	49.58	57.71	62.16	66.65	73.19	0.578	2

TIPSv2 3.0 to 1.7, SigLIP2 2.6 to 0.7, RADIO 0.7 to −0.6), so annotations mostly demote the correct part from first place to a nearby rank rather than losing it. The exchange is one-sided only for DINOv3, which loses 1,132 first-place hits and gains 428; RADIO loses 862 and gains 795, a wash. Clean-drawing results for all cutoffs are in Tab. 11.

C. Vision-Language Model Results

C.1. Inference Settings

Tab. 13 lists the model identifiers and decoding settings. Closed models were queried through their vendors' APIs between 15 and 25 August 2026. No temperature, sampling, or reasoning parameter was set in any call, so each closed model ran with its provider's defaults, and the identifiers are the aliases passed to the API, which the vendor resolves to its snapshot current at that time. The zero-shot open models were served through an OpenAI-compatible local endpoint using the "vllm" library, and likewise ran with the server's default sampling, taken from each checkpoint's generation config. The fine-tuned model ran in-process with Hugging Face Transformers in bf16 with greedy decoding, so its comparison with zero-shot Qwen3-VL also changes the decoding. Every image is stored at 512×512 px and sent to every model at that resolution, in the order given by each prompt. Responses without a parseable answer are excluded from that model's accuracy, and every paired comparison uses only the items both systems answered. The forced-choice runs contain one empty GPT-5.6 Luna response (Tab. 17), and the CoT runs lose 41 GPT-5.6 Luna responses (empty), 11 Gemini responses (API errors), and 3 Claude responses (empty or cut off at the token limit); Tab. 7 reports the remaining n. The same rule removes 52 GPT-5.6 Luna and 7 Claude empty responses and 6 Gemini API errors from Task 2, and 4 Gemini API errors and 4 unparseable Mistral-S answers from Task 3; every other Task 2 and 3 run answered all 1,000 items. The reasoning setting matters for reading Tab. 7: a model with internal reasoning enabled already reasons before its forced-choice answer, so for it the CoT control changes only whether that reasoning is written out and guided.

C.2. Paired Significance Tests

All VLMs answer the same items, so accuracy differences are tested on paired outcomes rather than by overlapping per-model intervals. For two systems A and B, let b be the items A answers correctly and B does not, and c the reverse; the exact (binomial) McNemar test asks whether b and c differ, using only the $b + c$ discordant items, and $\Delta\text{Acc} = (b - c)/n$. Each comparison uses the n items both systems answered (Sec. C.1), so ΔAcc can differ from the difference of Tab. 4 entries by 0.001–0.002. p-values are Holm-corrected within each block of Tab. 14; the leading-model blocks list the four highest-accuracy zero-shot models of each condition. Over all 15 zero-shot pairs per condition, Mistral-S is below every other model ($p_{\text{Holm}} < 0.001$) except Claude with dimensions ($p_{\text{Holm}} = 0.089$), and with dimensions Claude is below Gemini, GPT-5.6, and Qwen3-VL ($p_{\text{Holm}} \leq 0.010$); no other pair differs.

C.3. Task 1 by Surface-Type Class and Tier

Tab. 15 splits Task 1 by surface-type class and pair tier. Every model, the fine-tuned one included, scores 2–13 points higher on freeform than on prismatic parts. The tiers expose the response biases: Gemini, InternVL3.5, and Mistral-S reject negatives at 0.82–0.99 but accept positives at only 0.09–0.42, while Claude and GPT-5.6 show the reverse, most strongly with dimensions (Claude: 0.85 on positives, 0.33 on negatives). Same-class negatives stay within 0.07 of any-class negatives for all models, including the fine-tuned one despite its silhouette-mined training negatives (Sec. D); since they share only the coarse class, not overall shape, this does not test look-alike parts.

C.4. Task 1: Render Viewpoint

Tabs. 16 and 17 split Task 1 by render viewpoint. Positive-pair accuracy is highest on the side renders and, for every model but Gemini (lowest on *back*), lowest on *top* and *bottom*, where negative-pair accuracy is often higher than elsewhere (Claude: 0.72 on *top* against 0.48–0.49 on the side views without dimensions), so part of the drop on positives is a shift toward *No* on these views.

C.5. Task 2 by Surface-Type Class

Tab. 18 splits Task 2 by surface-type class. Claude, Gemini, GPT-5.6, and Mistral-S do better on freeform parts, InternVL3.5 and Qwen3-VL marginally better on prismatic ones. Against chance (one-sided exact binomial test, Holm-corrected over the six zero-shot models), Gemini, GPT-5.6, Claude, and Qwen3-VL score above 0.25 ($p_{\text{Holm}} < 10^{-5}$) and InternVL3.5 and Mistral-S do not ($p_{\text{Holm}} \geq 0.088$);

Table 12. **Per-view analysis of the four encoders**. Drawing→render, view-aligned mean pooling, gallery N=10,000. Single orthographic views (top block, sorted by R@1) and two-view fusion (bottom block). Best pair per backbone and metric in bold.

DINOv3

	View	R@1↑	R@5↑	R@10↑	MRR↑	MedR↓
Single	top	30.98	44.49	50.60	0.377	10
	bottom	25.87	38.72	44.42	0.323	19
	left	16.53	28.20	33.95	0.225	50
	right	16.41	27.58	33.40	0.223	55
	front	14.88	25.98	31.38	0.206	62
	back	14.49	24.26	29.54	0.198	72
Pairs	top + left	**53.01**	**72.35**	**78.75**	**0.621**	**1**
	front + top	51.11	70.74	77.20	0.602	**1**
	top + bottom	32.99	47.39	53.52	0.402	7
	front + left	29.05	44.30	51.03	0.365	9
	left + right	18.24	30.24	36.24	0.244	38
	front + back	15.76	27.08	33.11	0.217	48

TIPSv2

	View	R@1↑	R@5↑	R@10↑	MRR↑	MedR↓
Single	top	32.58	48.31	54.39	0.402	6
	bottom	27.56	41.96	48.53	0.347	12
	left	15.71	27.10	32.36	0.216	58
	right	15.66	27.07	32.14	0.215	57
	front	13.32	24.14	30.18	0.191	71
	back	12.79	23.12	28.70	0.183	80
Pairs	top + left	**54.22**	**74.20**	**80.66**	**0.634**	**1**
	front + top	50.70	71.53	78.38	0.603	**1**
	top + bottom	34.70	50.16	56.69	0.423	5
	front + left	27.65	42.23	49.05	0.349	12
	left + right	16.57	28.46	34.52	0.228	44
	front + back	13.96	25.14	31.24	0.199	59

SigLIP2

	View	R@1↑	R@5↑	R@10↑	MRR↑	MedR↓
Single	top	30.82	46.06	52.32	0.383	8
	bottom	23.68	37.42	43.49	0.305	20
	left	15.02	26.36	32.30	0.210	54
	right	14.85	26.62	32.44	0.209	53
	front	13.85	24.82	30.46	0.195	67
	back	12.53	23.49	29.55	0.183	76
Pairs	top + left	**51.16**	**70.75**	**77.36**	**0.603**	**1**
	front + top	49.83	69.10	76.10	0.589	2
	top + bottom	33.32	49.40	56.22	0.412	6
	front + left	28.22	43.72	50.34	0.358	10
	left + right	16.19	28.70	35.38	0.227	40
	front + back	14.35	26.60	32.59	0.206	54

RADIO

	View	R@1↑	R@5↑	R@10↑	MRR↑	MedR↓
Single	top	29.95	44.06	50.42	0.370	10
	bottom	23.34	36.07	42.19	0.297	25
	left	13.85	24.63	30.71	0.195	69
	right	13.44	24.41	30.26	0.192	69
	front	11.42	22.02	27.51	0.170	84
	back	10.96	21.42	26.53	0.164	96
Pairs	top + left	**48.23**	**67.78**	**74.91**	**0.575**	**2**
	front + top	45.62	66.28	73.30	0.552	**2**
	top + bottom	29.96	44.85	51.52	0.374	9
	front + left	25.23	40.70	46.57	0.326	15
	left + right	14.28	25.77	31.85	0.203	54
	front + back	11.93	23.02	28.71	0.178	71

Table 13. **VLM inference settings.** Identifier: API model alias (closed, queried 15–25 August 2026) or Hugging Face checkpoint (open). Temp.: no run set a temperature, so "default" is the provider's or the serving engine's default. Reasoning: "off" if thinking is disabled, "default" if left unset, n/a for models without a reasoning mode. Max tok.: output-token limit passed in the call, the same for forced-choice and CoT prompts; for GPT-5.6 Luna it also counts reasoning tokens.

Model	Identifier	Temp.	Reasoning	Max tok.
Claude Sonnet 5	`claude-sonnet-5`	default	off	2000
Gemini 3.6 Flash	`gemini-3.6-flash`	default	default	2000
GPT-5.6 Luna	`gpt-5.6-luna`	default	default	2000
Mistral-S 3.2 24B	`mistralai/Mistral-Small-3.2-24B-Instruct-2506`	default	n/a	2000 (20 in T3)
InternVL3.5 14B	`OpenGVLab/InternVL3_5-14B`	default	off	2000 (100 in T3)
Qwen3-VL 8B	`Qwen/Qwen3-VL-8B-Instruct`	default	n/a	2000
Qwen3-VL 8B (FT)	base + merged LoRA	greedy	n/a	2000

on the items both answered, Gemini leads GPT-5.6 by 6.3 points (paired McNemar $p < 0.001$). The fine-tuned Qwen3-VL drops to chance on both classes (overall 0.241, $p = 0.53$ against 0.25), 7.6 points below its zero-shot base ($p_{\text{Holm}} < 0.001$; Tab. 14): it treats the drawing as the first labelled candidate, answering *E* on 12.9% of items and the letter one past the ground truth on 54.1%.

C.6. Task 3: Complexity Probe

On a balanced set, every model but GPT-5.6 Luna (46.4%) answers *Simple* on 87.6–98.4% of items (Claude 93.4%, Gemini 87.6%, Mistral-S 93.4%, InternVL3.5 92.2%, Qwen3-VL 98.4%). Fine-tuning on Task 1 moves Qwen3-VL from 98.4% to 84.9% *Simple* answers (accuracy 0.512 to 0.553). Fig. 8 shows qualitative samples.

C.7. Task 1 Failure Analysis

Tab. 19 groups Task 1 items by model consensus. Models disagree on ∼80% of items (86–95% of positives); negatives are almost never failed unanimously ($< 1\%$) and are solved by all models in 14–41% of cases, whereas positives are solved unanimously in only 0.4–3.7%. Figs. 9–11 give examples.

D. Fine-Tuning Splits and Pair Construction

Data and settings used to adapt Qwen3-VL 8B to Task 1; all steps are deterministic given the released seed (13) and scripts.

Table 14. **Paired exact McNemar tests** on the items both systems answered. ΔAcc: accuracy of the first system minus the second on those items; $b+c$: discordant items; p_{Holm}: Holm-corrected within each block.

Comparison	ΔAcc	$b+c$	p_{Holm}
Task 1, leading models (undimensioned)			
GPT-5.6 vs. Qwen3-VL	+0.000	342	1.000
GPT-5.6 vs. Gemini	+0.017	431	1.000
GPT-5.6 vs. Claude	+0.036	344	0.295
Qwen3-VL vs. Gemini	+0.016	418	1.000
Qwen3-VL vs. Claude	+0.035	295	0.285
Gemini vs. Claude	+0.019	497	1.000
Task 1, leading models (dimensioned)			
Gemini vs. Qwen3-VL	+0.010	378	1.000
Gemini vs. GPT-5.6	+0.016	470	1.000
Gemini vs. InternVL3.5	+0.033	307	0.406
Qwen3-VL vs. GPT-5.6	+0.006	400	1.000
Qwen3-VL vs. InternVL3.5	+0.023	277	0.931
GPT-5.6 vs. InternVL3.5	+0.017	443	1.000
Task 1, dimensioned vs. undimensioned			
Claude Sonnet 5	−0.045	227	0.020
Gemini 3.6 Flash	+0.016	148	1.000
GPT-5.6 Luna	−0.016	254	1.000
Mistral-S 3.2 24B	−0.010	124	1.000
InternVL3.5 14B	+0.013	217	1.000
Qwen3-VL 8B	−0.010	148	1.000
Task 1, CoT vs. forced choice (undimensioned)			
Claude Sonnet 5	+0.038	302	0.165
Gemini 3.6 Flash	+0.014	134	1.000
GPT-5.6 Luna	+0.007	277	1.000
Mistral-S 3.2 24B	+0.011	673	1.000
InternVL3.5 14B	−0.007	195	1.000
Qwen3-VL 8B	−0.049	221	0.007
Qwen3-VL 8B, fine-tuned vs. zero-shot (Task 1)			
Task 1	+0.016	300	0.387
Task 1 + DIM	+0.040	260	0.031
Qwen3-VL 8B, fine-tuned vs. zero-shot (Tasks 2 and 3)			
Task 2	−0.076	314	< 0.001
Task 3	+0.041	143	< 0.001

D.1. Splits

The test split is the VLM evaluation set (Sec. 4.2); the remaining models are split by identifier into training and validation (5%), and every pair is built within one split (Tab. 20).

D.2. Render View Labels

Each render's view is read from its camera pose by mapping the dominant axis to the projection directions of Tab. 10, and verified on 200 models against the drawings by silhouette IoU. Every render matches its own axis best (mean IoU 0.85–0.92 against 0.31–0.84 for other views).

D.3. Hard Negatives

Hard negatives come from a model-free silhouette descriptor (six 16×16 view masks plus aspect ratios), so they share the outline and differ in interior structure such as holes, steps, and hidden lines; a nearest neighbour is far closer than a random model (cosine 0.85 vs. 0.01).

Table 15. Task 1 VLM results by surface-type class and pair tier, without (left) and with (right) dimension annotations. Bold: best zero-shot model per column; the fine-tuned (FT) row is excluded.

		Without DIM				With DIM			
	Model	Acc.	F1	Rec.	Prec.	Acc.	F1	Rec.	Prec.
Complex	Claude Sonnet 5	0.665	0.665	0.666	0.668	0.611	0.597	0.615	0.637
	Gemini 3.6 Flash	**0.717**	0.699	**0.713**	**0.777**	**0.735**	**0.724**	**0.732**	**0.773**
	GPT-5.6 Luna	0.711	**0.710**	0.710	0.711	0.705	0.704	0.707	0.711
	Mistral-S 3.2 24B	0.545	0.428	0.537	0.675	0.563	0.488	0.556	0.628
	InternVL3.5 14B	0.659	0.638	0.655	0.698	0.663	0.639	0.659	0.711
	Qwen3-VL 8B	0.679	0.668	0.676	0.702	0.689	0.677	0.686	0.715
	Qwen3-VL 8B (FT)	0.712	0.694	0.707	0.763	0.716	0.706	0.713	0.742
Simple	Claude Sonnet 5	0.604	0.595	0.601	0.610	0.568	0.516	0.562	0.604
	Gemini 3.6 Flash	0.590	0.534	0.597	**0.697**	0.604	0.561	0.610	**0.689**
	GPT-5.6 Luna	0.630	0.626	0.628	0.633	0.602	0.589	0.599	0.611
	Mistral-S 3.2 24B	0.522	0.406	0.531	0.670	0.484	0.413	0.491	0.482
	InternVL3.5 14B	0.588	0.570	0.592	0.613	0.610	0.597	0.614	0.634
	Qwen3-VL 8B	**0.660**	**0.658**	**0.659**	0.661	**0.630**	**0.629**	**0.631**	0.633
	Qwen3-VL 8B (FT)	0.658	0.635	0.663	0.724	0.682	0.670	0.686	0.723
		Same-cl. n.	**Any-cl. n.**	**Pos.**		**Same-cl. n.**	**Any-cl. n.**	**Pos.**	
Tier acc.	Claude Sonnet 5	0.562	0.520	**0.728**		0.329	0.332	**0.848**	
	Gemini 3.6 Flash	0.952	0.952	0.356		**0.924**	**0.932**	0.412	
	GPT-5.6 Luna	0.655	0.620	0.703		0.554	0.516	0.772	
	Mistral-S 3.2 24B	**0.976**	**0.988**	0.086		0.876	0.904	0.158	
	InternVL3.5 14B	0.815	0.880	0.400		0.863	0.852	0.416	
	Qwen3-VL 8B	0.695	0.764	0.610		0.767	0.792	0.540	
	Qwen3-VL 8B (FT)	0.924	0.936	0.440		0.856	0.920	0.510	

Table 16. Task 1 accuracy by render viewpoint, with DIM, under the forced-choice Task 1 prompt of Sec. 5.2 (weighting by n recovers the positive and negative accuracies of Tab. 6). Bold: best viewpoint per model; the fine-tuned (FT) row is not marked.

Model	front	right	back	left	top	bottom
Positive pairs (n = 86, 81, 92, 79, 78, 84)						
Claude Sonnet 5	0.92	**1.00**	0.89	**1.00**	0.53	0.74
Gemini 3.6 Flash	0.35	**0.56**	0.27	0.54	0.42	0.36
GPT-5.6 Luna	0.76	**0.95**	0.74	0.89	0.63	0.68
Mistral-S 3.2 24B	0.16	0.21	0.20	**0.28**	0.06	0.04
InternVL3.5 14B	0.43	**0.68**	0.42	0.66	0.14	0.17
Qwen3-VL 8B	0.55	0.77	0.60	**0.89**	0.19	0.25
Qwen3-VL 8B (FT)	0.49	0.78	0.51	0.81	0.22	0.26
Negative pairs (n = 73, 89, 85, 84, 89, 79)						
Claude Sonnet 5	0.30	0.29	0.31	0.24	**0.44**	0.41
Gemini 3.6 Flash	**0.95**	0.93	0.93	0.92	0.94	0.90
GPT-5.6 Luna	0.52	0.56	**0.61**	0.40	0.54	0.57
Mistral-S 3.2 24B	0.84	0.88	0.86	0.87	0.92	**0.97**
InternVL3.5 14B	0.82	0.83	0.84	0.83	**0.91**	**0.91**
Qwen3-VL 8B	0.82	0.72	0.74	0.67	**0.89**	0.85
Qwen3-VL 8B (FT)	0.86	0.94	0.93	0.88	0.84	0.86

D.4. Pairs

Each model contributes two clean drawing queries from two of its six views, each with one positive and one negative; 70% of negatives are drawn from the three nearest silhouette neighbours, the rest at random, and the render view is sampled independently of whether the pair is positive or negative (Tab. 20).

Table 17. Task 1 accuracy by render viewpoint, without DIM, under the forced-choice Task 1 prompt of Sec. 5.2 (weighting by n recovers the positive and negative accuracies of Tab. 6). Bold: best viewpoint per model; the fine-tuned (FT) row is not marked.

Model	front	right	back	left	top	bottom
Positive pairs (n = 86, 81, 92, 79, 78, 84/83†)						
Claude Sonnet 5	0.77	**0.98**	0.68	0.97	0.46	0.51
Gemini 3.6 Flash	0.31	**0.47**	0.18	**0.47**	0.42	0.31
GPT-5.6 Luna	0.74	**0.88**	0.67	0.85	0.51	0.57
Mistral-S 3.2 24B	**0.12**	**0.12**	0.10	0.11	0.06	0.00
InternVL3.5 14B	0.41	0.65	0.39	**0.71**	0.12	0.13
Qwen3-VL 8B	0.66	0.89	0.67	**0.91**	0.21	0.31
Qwen3-VL 8B (FT)	0.37	0.70	0.47	0.77	0.14	0.19
Negative pairs (n = 73, 89, 85, 84, 89, 79)						
Claude Sonnet 5	0.48	0.49	0.49	0.48	**0.72**	0.57
Gemini 3.6 Flash	0.97	0.96	0.96	**0.98**	0.93	0.91
GPT-5.6 Luna	0.58	**0.70**	0.69	0.57	0.64	0.63
Mistral-S 3.2 24B	0.99	0.99	0.95	0.98	**1.00**	0.99
InternVL3.5 14B	0.89	0.85	0.82	0.74	**0.90**	0.89
Qwen3-VL 8B	0.70	0.69	0.73	0.62	**0.83**	0.81
Qwen3-VL 8B (FT)	0.95	0.94	0.92	0.90	0.90	0.97

†GPT-5.6 Luna: n=83 for *bottom* positives due to one empty response.

Table 18. Task 2 VLM results by model (four-way drawing-to-render retrieval). Bold: best zero-shot model per column; the fine-tuned (FT) row is excluded

	Model	Acc.	F1	Rec.	Prec.
Overall	Claude Sonnet 5	0.374	0.372	0.379	0.425
	Gemini 3.6 Flash	**0.690**	**0.689**	**0.689**	**0.691**
	GPT-5.6 Luna	0.637	0.636	0.635	0.643
	Mistral-S 3.2 24B	0.274	0.235	0.267	0.282
	InternVL3.5 14B	0.266	0.274	0.269	0.310
	Qwen3-VL 8B	0.317	0.305	0.321	0.322
	Qwen3-VL 8B (FT)	0.241	0.246	0.239	0.257
Complex	Claude Sonnet 5	0.415	0.415	0.413	0.459
	Gemini 3.6 Flash	**0.817**	**0.817**	**0.816**	**0.817**
	GPT-5.6 Luna	0.740	0.740	0.739	0.744
	Mistral-S 3.2 24B	0.284	0.250	0.285	0.288
	InternVL3.5 14B	0.264	0.276	0.264	0.309
	Qwen3-VL 8B	0.316	0.307	0.317	0.322
	Qwen3-VL 8B (FT)	0.236	0.241	0.233	0.256
Simple	Claude Sonnet 5	0.332	0.326	0.343	0.389
	Gemini 3.6 Flash	**0.564**	**0.563**	**0.563**	**0.566**
	GPT-5.6 Luna	0.528	0.521	0.522	0.531
	Mistral-S 3.2 24B	0.264	0.221	0.250	0.293
	InternVL3.5 14B	0.268	0.274	0.274	0.320
	Qwen3-VL 8B	0.318	0.303	0.325	0.325
	Qwen3-VL 8B (FT)	0.246	0.250	0.244	0.260

Test pairs are the Task 1 items of the main text (1,000 test models, front-view drawing; 500 positives, 250 any-class and 250 same-class negatives), in clean and dimensioned form, so zero-shot and fine-tuned rows are scored on identical pairs.



Figure 7. Examples of 2D drawing retrieval tasks with four candidate CAD renders. Green border: ground truth; blue border: model selection.

Table 19. Task 1 failure analysis by surface-type class, negative type, and prediction outcome.

Group	Tier	Result	Count (%)
Complex	Any-cl. neg. (n=122)	All success	50 (41.0%)
		Mixed	72 (59.0%)
	Same-cl. neg. (n=133)	All failed	1 (0.8%)
		All success	53 (39.8%)
		Mixed	79 (59.4%)
	Positive (n=245)	All failed	25 (10.2%)
		All success	9 (3.7%)
		Mixed	211 (86.1%)
Simple	Any-cl. neg. (n=128)	All success	27 (21.1%)
		Mixed	101 (78.9%)
	Same-cl. neg. (n=117)	All success	17 (14.5%)
		Mixed	100 (85.5%)
	Positive (n=255)	All failed	13 (5.1%)
		All success	1 (0.4%)
		Mixed	241 (94.5%)



(a) Open-source models.

(b) Closed-source models.

Figure 8. Task 3 predictions across models (green = correct, red = incorrect).

Table 20. **Fine-tuning pairs.** Same view: share of positive / negative pairs whose render shows the drawing's view.

Split	Models	Pairs	Pos.	Hard neg.	Same view
Train	27,479	109,916	54,958	38,483	0.166 / 0.166
Validation	1,446	5,784	2,892	2,058	0.169 / 0.162

D.5. Training

LoRA [27] adapters cover the language model and the vision encoder, the base model is held in 4-bit during training, and the loss covers only the answer tokens. Tab. 21 gives the training prompt, which places the drawing first and the render second, and the full configuration.

D.6. Evaluation

All *Qwen3-VL 8B (FT)* numbers come from the zero-shot harness: the Task 1–3 prompts of Sec. 5.2, the same item sets, exact string matching, greedy decoding (rather than the default sampling of the zero-shot open models, Tab. 13), and the LoRA weights merged into the base model after one epoch (1,250 steps at an effective batch of 16, final training loss 0.19). Tasks 2 and 3 were never seen in training and measure transfer. Under the chain-of-thought prompt the model answered with a single word on every item and reproduced its forced-choice predictions exactly, so Table 7 repeats those numbers.

D.7. Caveats

Hard negatives are selected by outline alone, so they test whether a model reads interior detail rather than whether it separates parts with different outlines but matching local features; and all splits come from one generator, so gains measure in-distribution adaptation (Sec. 7).

Table 21. **Fine-tuning prompt and configuration** (Qwen3-VL 8B). The prompt is used for training only; evaluation uses the Task 1 prompt of the main text.

VLM Prompt. *You are shown two images of a mechanical part. The first image is a 2D engineering drawing. The second image is a shaded render of a 3D CAD model. The two images may show the part from different viewpoints. Do they depict the same part? Answer with exactly one word: Yes or No.*

Setting	Value
LoRA rank / α / dropout	16 / 32 / 0.05
Trainable parameters	52.5M (0.60%)
LoRA targets (LM)	`q,k,v,o,gate,up,down_proj`
LoRA targets (vision)	`qkv,proj,linear_fc1,linear_fc2`
Optimizer	AdamW, lr 10^{-4}, cosine schedule, 3% warmup
Effective batch	16 (1 pair $\times$ 16 accumulation)
Training pairs	20,000, 1 epoch
Validation	500 pairs every 250 steps, best loss kept
Base model	4-bit NF4 (QLoRA [18]); vision tower, LM head bf16
Images	512×512 px, 256 tokens each
Hardware	$1\times$ L4 GPU (24 GB)

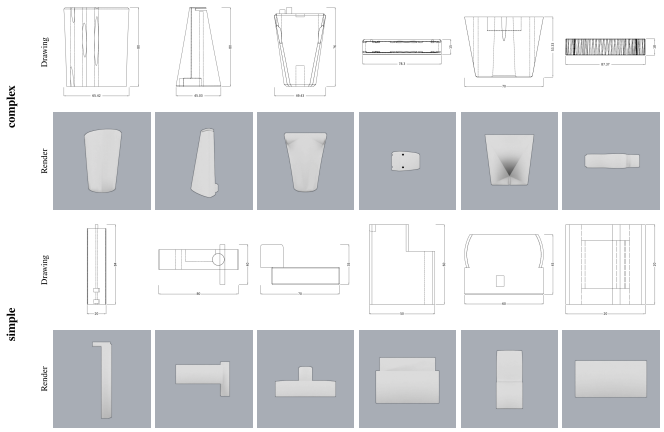


Figure 9. Task 1 examples where all evaluated VLMs misclassified positive pairs.

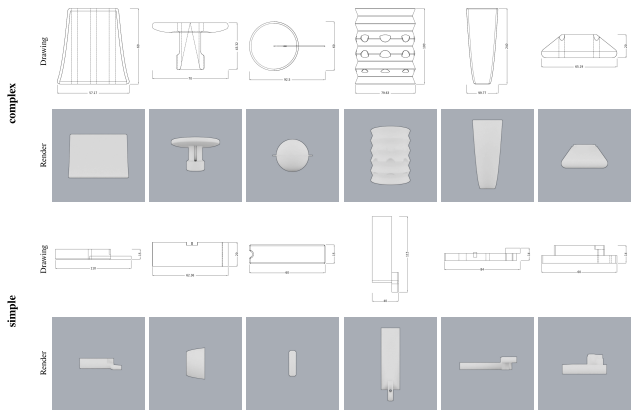


Figure 10. Task 1 examples where all evaluated VLMs correctly matched pairs.

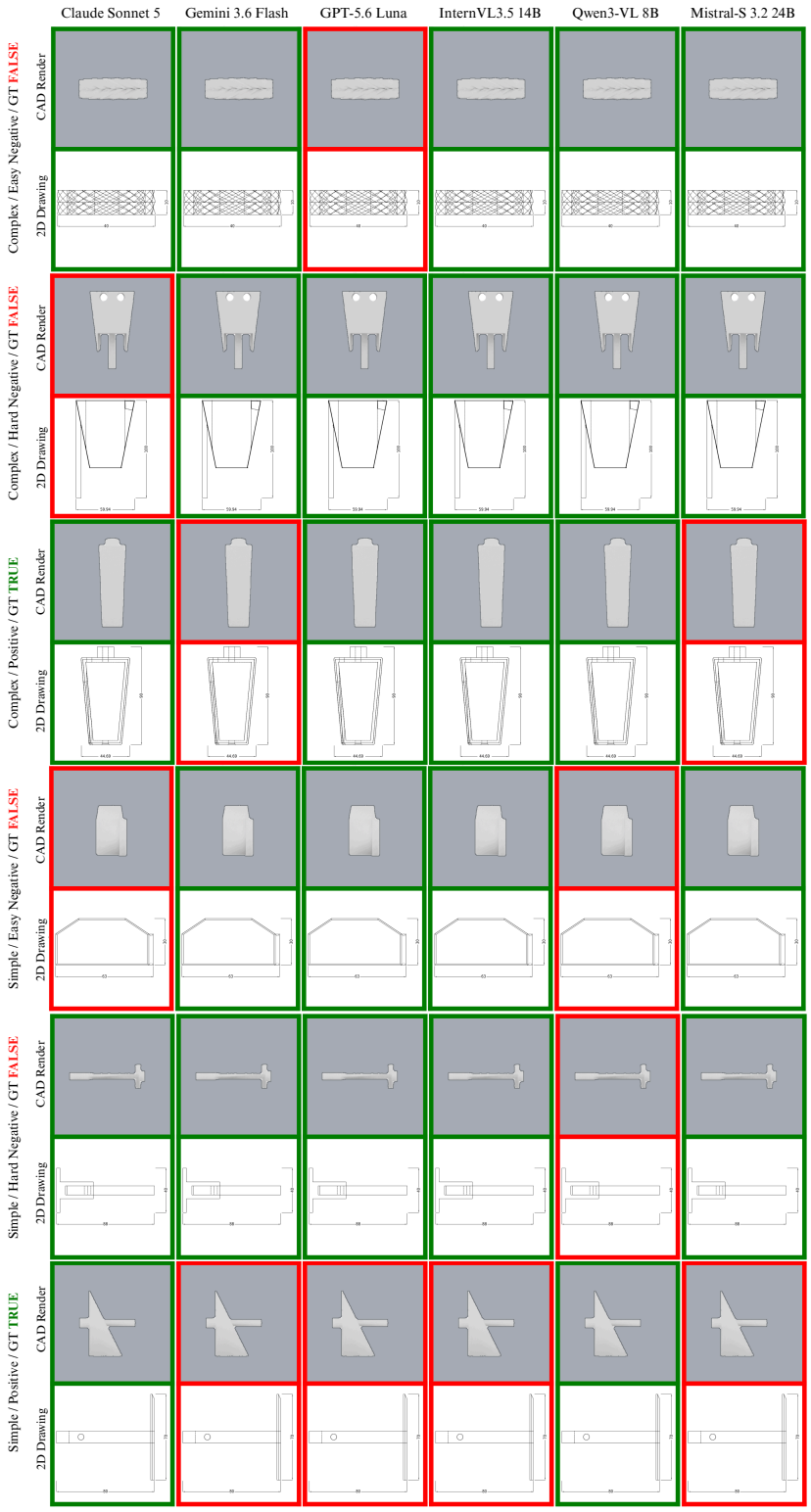


Figure 11. Task 1 qualitative examples showing mixed model predictions. Rows labelled *Hard Negative* are same-class negatives (same prismatic/freeform class) and *Easy Negative* rows are any-class negatives; neither is mined for visual similarity.